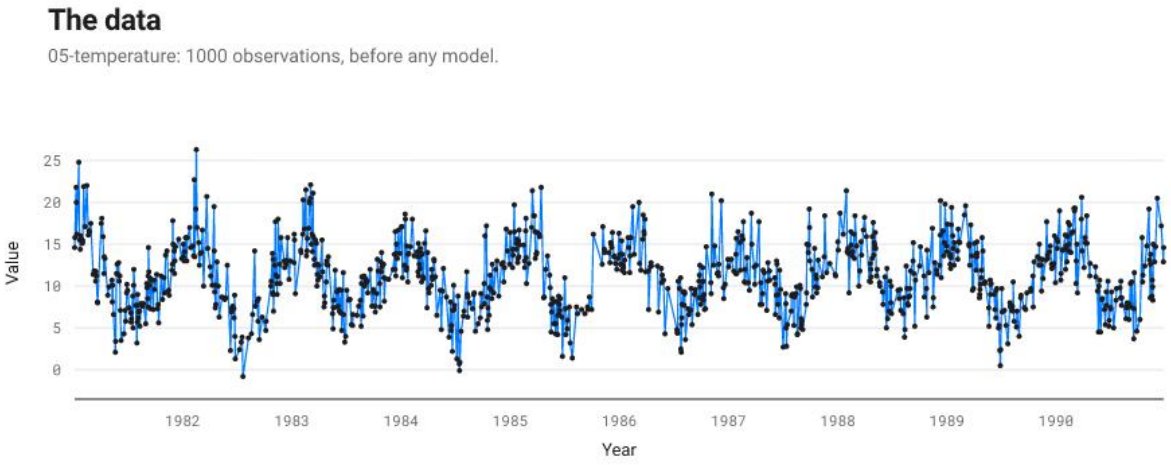
An automatic report for the dataset: 05-temperature

Some Math empowered by some AI guided by some researchers

09 November 2026

Produced by ***autostat****, a Python 3 port of the Automatic Bayesian Covariance Discovery system of Lloyd et al. (AAAI 2014).*



The data

Executive summary

The structure search selected `C + WN + Per + SE` (Bayesian information criterion 4828.7; negative log likelihood 2390.2 on 1000 points). It decomposes into 4 additive components, listed by how much each reduces extrapolation error:

1. A constant.
2. A periodic function with a period of 1.0 years.
3. A rapidly varying smooth function.
4. Uncorrelated noise.

#	R^2 cumulative (%)	Residual variance explained (%)	CV MAE	MAE reduction (%)
data			11.2	
1	0.0	0.0	3.34	70.2
2	54.6	54.6	2.23	33.4
3	97.8	95.1	2.22	0.1
4	100.0	100.0	2.22	0.0

MAE: mean absolute error over ten contiguous held-out blocks; CV: cross-validated.

The formula found

The structure search is a symbolic regression: it searches a grammar of expressions over kernels and keeps the one with the lowest Bayesian information criterion. This is the expression it kept, with every fitted parameter in place; t is in years.

$$y(t) = f_1(t) + f_2(t) + f_3(t) + f_4(t)$$

Each component is a Gaussian process with its own covariance:

$$f_1(t) \sim \mathrm{GP}(0,\ \ 147.7)$$

$$f_2(t) \sim \mathrm{GP}\left(0,\ \ 7.967\,\frac{\exp\left(0.1033\cos\frac{2\pi(t-t')}{1.001}\right) - I_0(0.1033)}{e^{0.1033} - I_0(0.1033)}\right)$$

$$f_3(t) \sim \mathrm{GP}\left(0,\ \ 6.432\,\exp\left(-\frac{(t-t')^2}{2\cdot 0.003448^2}\right)\right)$$

$$f_4(t) \sim \mathrm{GP}(0,\ \ 1.459\,\delta_{tt'})$$

These components have a closed form, read from their posterior mean:

$$f_1(t) = 11.15$$

$$\begin{aligned} f_2(t) \approx\ & 4.293\cos\left(\frac{2\pi\cdot(t-1981)}{1.001} - 0.3861\right) + 0.5723\cos\left(\frac{2\pi\cdot 2\cdot(t-1981)}{1.001} - 2.754\right) \\ & + 0.009047\cos\left(\frac{2\pi\cdot 3\cdot(t-1981)}{1.001} + 1.623\right) \end{aligned}$$

A periodic kernel is exactly a sum of cosines, so a periodic component is a Fourier series in its period; it is written up to the harmonic after which the rest weighs less than 0.1

The other components are smooth, quasi-periodic or noise without a short formula: a draw from such a process has no finite closed form; its covariance above is the formula.

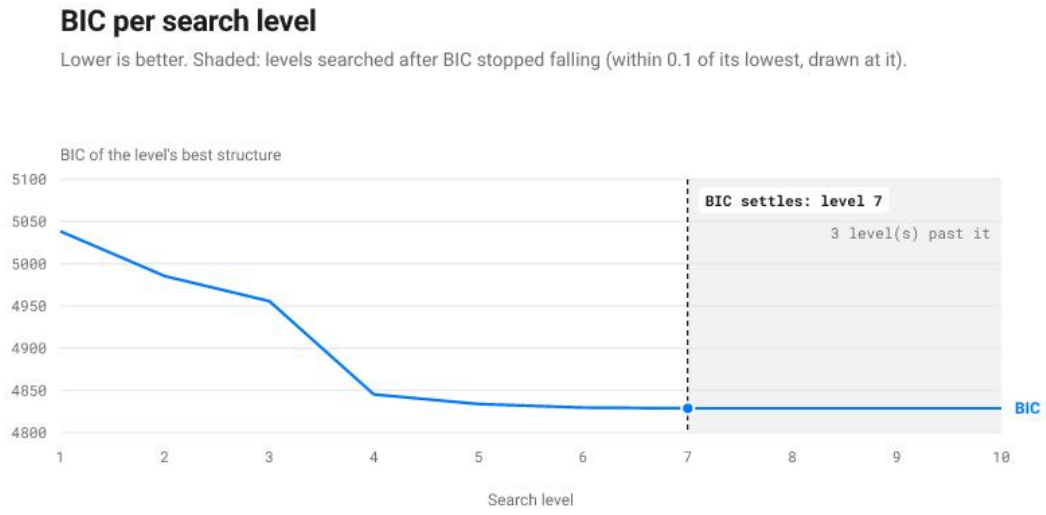
Closed form of each component:

#	component	closed form
1	A constant	exact polynomial of degree 0
2	A periodic function with a period of 1.0 years	Fourier series, 3 harmonics, largest gap 0.011 % of the component
3	A rapidly varying smooth function	none: the covariance is the formula
4	Uncorrelated noise	none: the covariance is the formula

Search path and stopping criterion

The 2014 configuration searches a fixed number of levels and keeps the lowest BIC; it has no stopping rule. The table follows the winner of each level. A change of structure adds or removes terms; a level that keeps the structure only refits its parameters. BIC differences read on the scale of Kass and Raftery (1995): under 2 barely worth mentioning, 2 to 6 positive, 6 to 10 strong, over 10 very strong.

level	BIC	change	structure	components	closed	params
1	5038.5		SE + WN	2	0	3
2	4985.2	-53.3	Per + SE + WN	3	1	6
3	4955.6	-29.6	Per + Per x SE + SE + SE + WN	5	1	12
4	4845.2	-110.4	Per + SE + SE + WN	4	1	8
5	4833.9	-11.3	C + Per + SE	3	2	6
6	4829.5	-4.4	C + Per + SE + WN	4	2	7
7	4828.8	-0.7	C + Per + SE + WN	4	2	7
8	4828.8	+0.0	C + Per + SE + WN	4	2	7
9 (kept)	4828.7	-0.1	C + Per + SE + WN	4	2	7
10	4828.7	+0.0	C + Per + SE + WN	4	2	7



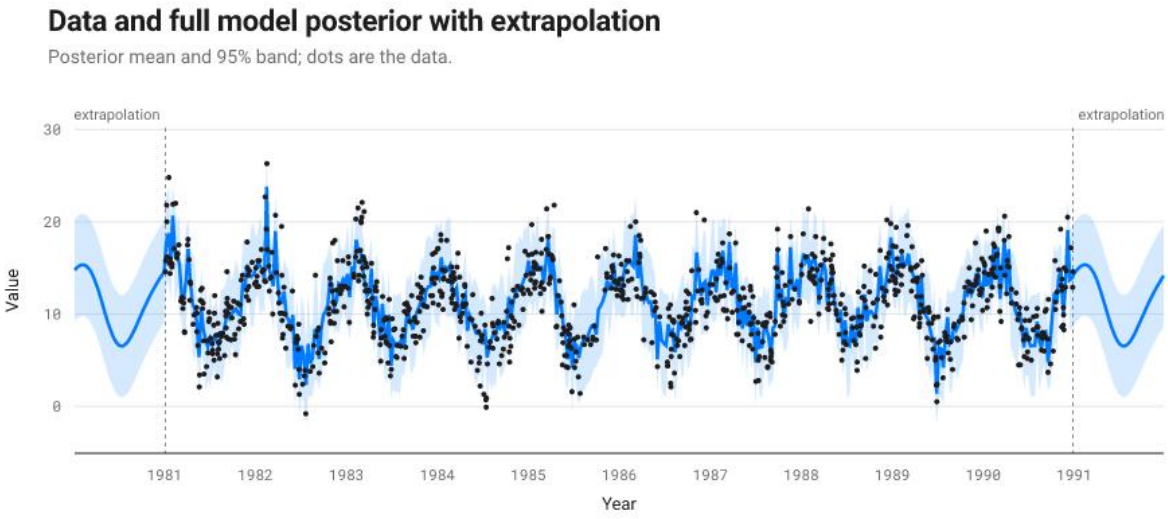
BIC per search level

Posterior weight of each structure met, exp(-BIC/2) normalised under a flat prior, after the Bayesian machine scientist (Guimerà et al., 2020): where the search keeps one structure, the weights say how much the data prefer it to the others. The 4 structure(s) under 0.001 are left out.

structure	lowest BIC	weight
C + Per + SE + WN	4828.7	0.931
C + Per + SE	4833.9	0.069

- The search ran 10 levels (the published 2014 search: ten levels, no stopping rule) and keeps the model of level 9, C + Per + SE + WN, BIC 4828.7.
- That structure was first reached at level 6; the levels after it only refitted its parameters, lowering BIC by 0.8 in all.
- The best different structure, C + Per + SE (level 5), is 5.2 above it: positive evidence for the kept one on the Kass and Raftery scale.
- Weighed by exp(-BIC/2) under a flat prior, as the Bayesian machine scientist weighs expressions, the structures met give the kept one a posterior weight of 0.931 and the next, C + Per + SE, 0.069: the data favour the kept one without ruling the next out.
- BIC stopped falling at level 7. The 3 level(s) searched past it kept its structure and lowered BIC by 0.09 in all, less than the tolerance of 0.1: going on buys nothing the data pay for.

- A rule stopping after 1 level(s) without a fall above 0.1 would have ended the search at level 8 and kept the same structure, less refitted.
- A rule stopping after 2 level(s) without a fall above 0.1 would have ended the search at level 9 and kept the same model.
- The kept expression has 7 parameters for 1000 points: its BIC is a fit term 2 NLL = 4780.3 plus a complexity term p ln n = 48.4, each extra parameter costing 6.91; it splits into 4 components, 2 of them with a closed form.



Data and full posterior

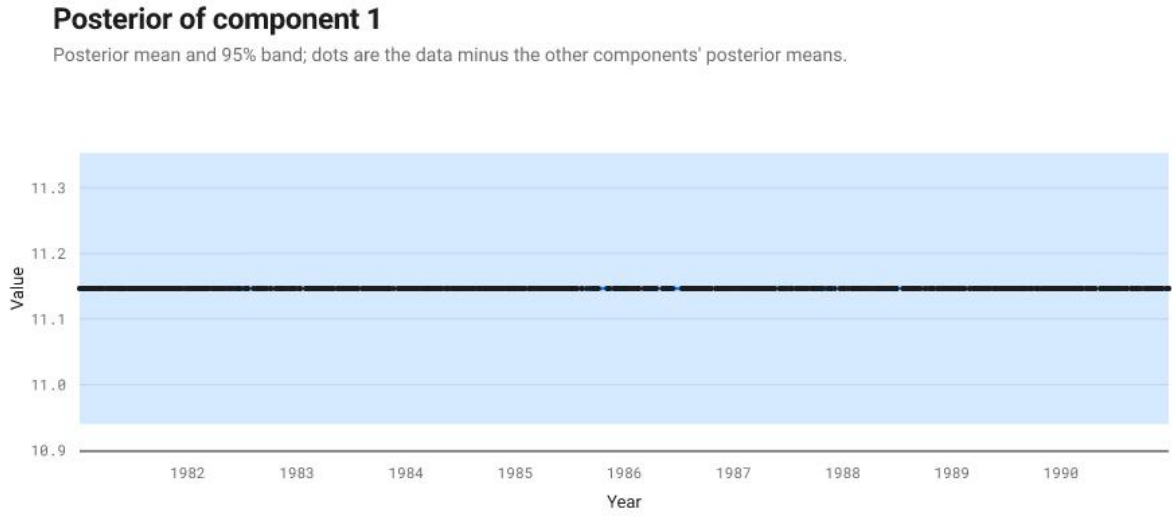
Detailed discussion of the components

Component 1: A constant

This component is constant.

This component explains 0.0% of the data's variance on its own; with the components above it, 0.0%. Adding it changes the cross-validated MAE by -70.2%.

Extrapolation. This component is assumed to stay constant.



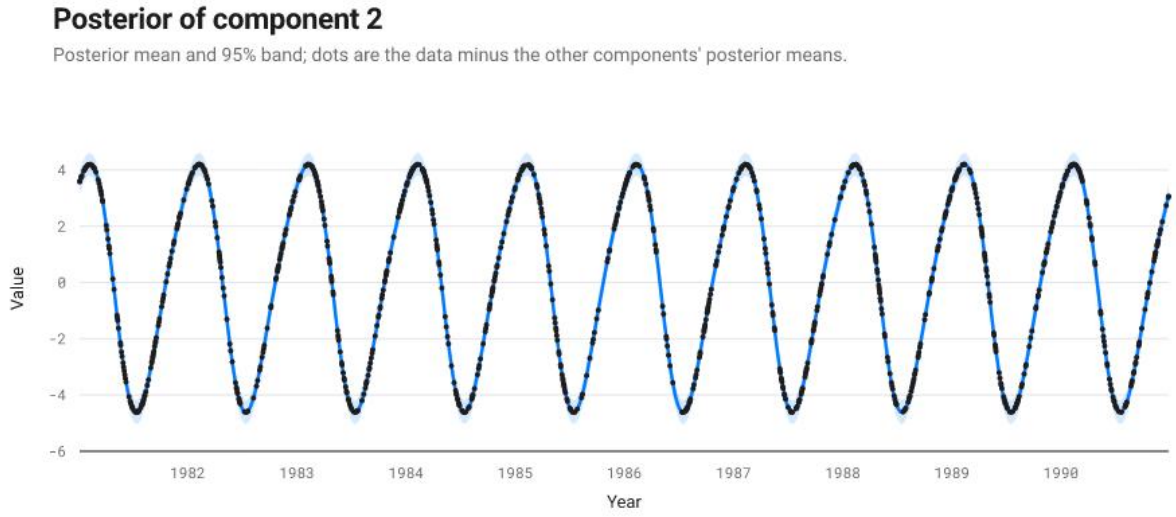
Component 1

Component 2: A periodic function with a period of 1.0 years

This component is periodic with a period of 1.0 years. The shape of this function within each period has a typical lengthscale of 5.9 months.

This component explains 54.6% of the data's variance on its own; with the components above it, 54.6%. Adding it changes the cross-validated MAE by -33.4%.

Extrapolation. This component is assumed to continue periodically with a period of 1.0 years.



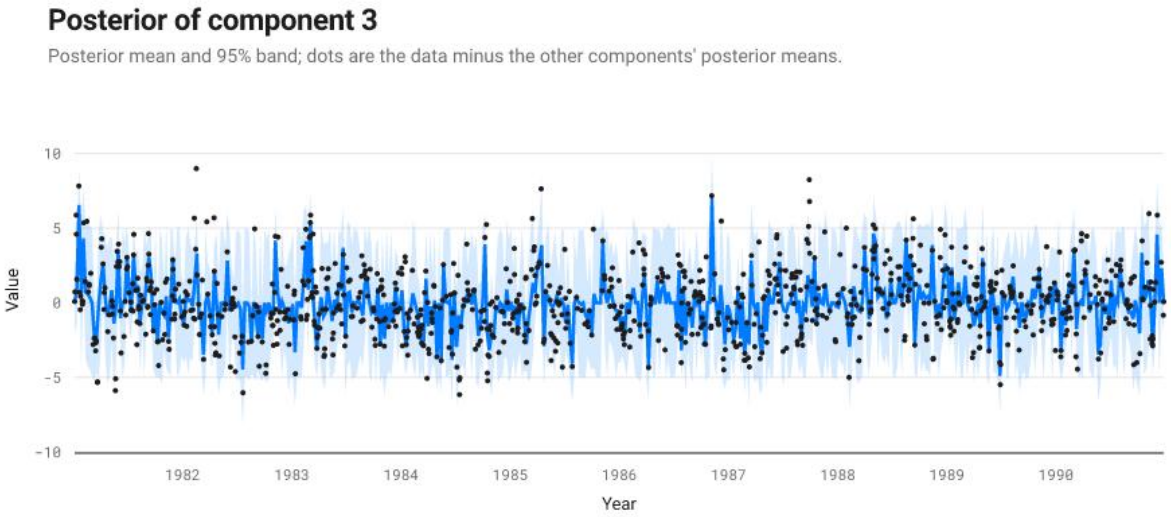
Component 2

Component 3: A rapidly varying smooth function

This component is a rapidly varying but smooth function with a typical lengthscale of 30.2 hours.

This component explains 41.8% of the data's variance on its own; with the components above it, 97.8%. Adding it changes the cross-validated MAE by -0.1%.

Extrapolation. This component is assumed to continue smoothly but its distribution is assumed to quickly return to the prior. The prior distribution places mass on smooth functions with a marginal mean of zero and a typical lengthscale of 30.2 hours.



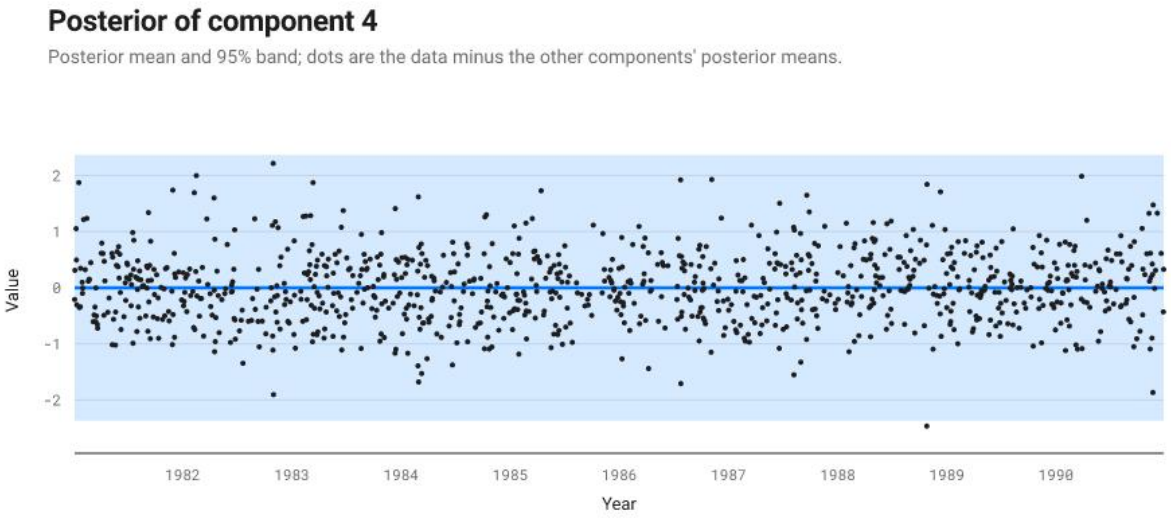
Component 3

Component 4: Uncorrelated noise

This component models uncorrelated noise.

This component explains 14.8% of the data's variance on its own; with the components above it, 100.0%. Adding it changes the cross-validated MAE by -0.0%.

Extrapolation. This component assumes the uncorrelated noise will continue indefinitely.



Component 4

Model checking

Each statistic compares draws of a component's posterior with draws of its prior. A small p-value (bold below 0.05) points at a property of the data that the component's prior does not

reproduce.

#	QQ max	QQ min	ACF min	ACF min lag	Pgram max	Pgram peak	MMD
1	0.330	0.330	0.992	0.954	0.728	0.966	**0.000**
2	0.508	0.680	0.876	0.460	0.646	0.990	**0.000**
3	0.214	0.280	0.906	0.772	0.894	0.734	**0.000**
4	0.834	0.896	0.992	0.990	0.954	0.998	**0.000**

QQ: quantile-quantile gaps. ACF: autocorrelation function. Pgram: periodogram. MMD: maximum mean discrepancy two-sample test.